Membership Circuits: Tractable Membership Testing via Probabilistic Circuits

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Abstract

Statistical hypothesis testing is a cornerstone of scientific progress, yet rigorous methods for high-dimensional data remain scarce. This gap severely limits reliable statistical inference in critical domains such as medicine and pharmaceutics. We address this limitation by introducing Membership Circuits (MCs), a model that leverages the structural properties of Probabilistic Circuits (PCs) to enable tractable, i.e., exact and efficient, computation of multivariate p-values, providing formal guarantees on the membership hypothesis. We present extensive evaluations of the test's properties and investigate its effectiveness in out-of-distribution (OOD) detection. Finally, we demonstrate that MCs are aware if the learned architecture fails to capture the original dataset.

1. Introduction

Statistical hypothesis testing serves as a fundamental scientific tool for data-driven decisions, as it offers probabilistic error control on well-defined questions in the presence of uncertainty (Fisher, 1932; Lehmann & Romano, 2005). A hypothesis test assesses a *hypothesis* about the relationship of sets of samples or distributions while bounding its *type I error rate*, i.e., the probability of falsely rejecting a true null hypothesis, or, in other words, the false positive rate (FPR). By bounding this error rate, statistical testing offers a level of reliability that is crucial in safety-critical domains like healthcare and yet is often missing in modern machine learning algorithms due to data's high dimensionality.

To illustrate the necessity of multivariate hypothesis testing, consider a diagnostic model trained on a "healthy" patient's data of high-dimensional biomarkers $\{x_1, \ldots, x_d\}$. A patient presents markers that, when viewed individually, are plausible w.r.t. the reference population. A series of univariate tests would fail to reject the null for any single dimension, concluding the sample is "normal". This conclusion might be incorrect, though. While each marker is marginally plausible, their joint occurrence may be highly implausible under the learned distribution $P(\boldsymbol{X})$. By neglecting feature dependencies, marginal tests fail to capture the true geometry of the data. Our framework addresses this by providing a multivariate membership test with formal False Positive Rate (FPR) control. In high-stakes settings, this allows for a rigorous bound on false alarms. Furthermore, if the empirical rejection rate significantly exceeds the user-defined threshold α, it serves as a mathematically grounded indicator of model failure, signaling that the underlying model has failed to capture the true distribution. Thus, our test is both a tool for out-of-distribution (OOD) detection and a critical diagnostic for model reliability.

	Classical Testing	Probabilistic Circuits	Membership Circuits
Bounding FPR	✓	✗	✓
Tractable	✗	✓	✓
Model mismatch	?	✗	✓

Table 1. Membership Circuits unify the probabilistic guarantees of statistical testing with the computational efficiency of Probabilistic Circuits. Moreover, they can identify when the underlying circuit does not model our data accurately.

Recent efforts have tried to bridge the gap between machine learning and statistical testing by either developing tests that assess properties of learned models (Dietterich, 1998; Nadeau & Bengio, 1999; Bouckaert & Frank, 2004; Nalisnick et al., 2019; Horel & Giesecke, 2020; Mandel & Barnett, 2024; Shi et al., 2024), or using machine learning techniques to compute or improve test statistics (Gretton et al., 2012; Lopez-Paz & Oquab, 2017; Liu et al., 2020; Kirchler et al., 2020). Another option is the construction of hypothesis tests around black-box models such as neural networks, but respective methods often come at the cost of auxiliary calibration datasets or expensive simulations to estimate test distributions (Nalisnick et al., 2019; Mandel & Barnett, 2024).

In this paper, we are interested in enabling statistical testing in learned multivariate distributions. Specifically, we are interested in membership testing, i.e., whether an observa-

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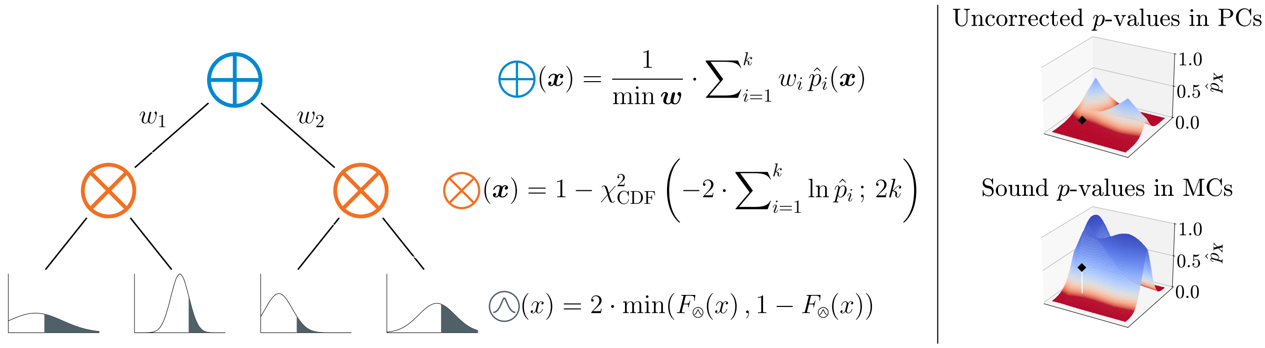

Figure 1. **Membership Circuits compute sound multivariate p-values by utilizing merging functions.** The tractable membership test is a single bottom-up inference pass that recursively merges p-values computed from the cumulative distribution functions F in the leaves $\oslash$. The merging functions ensure that the resulting multivariate p-values are sound (here: corrected arithmetic mean at sum nodes $\oplus$ (eq. 9), Fisher's method at product nodes $\otimes$ (eq. 12)).

tion $\boldsymbol{x}$ belongs to a learned distribution $P_{\boldsymbol{X}}$. Since testing in $P_{\boldsymbol{X}}$ directly is usually intractable, we utilize a surrogate distribution $P_{\mathcal{C}} \approx P_{\boldsymbol{X}}$ to test the null hypothesis

$$H_0 : \boldsymbol{x} \sim P_{\mathcal{C}}. \tag{1}$$

To achieve this, we introduce Membership Circuits (MCs) that exploit the explicit, hierarchical representation of joint probability distributions of Probabilistic Circuits (PCs) (Choi et al., 2020) and cast the problem of computing multivariate p-values as a multiple testing problem. MCs compute univariate p-values at leaf nodes and merge them at inner nodes. With this, tractability is preserved while enforcing the user-defined bound on the acceptable false positive rate. Tab. 1 summarizes the advantages of MCs compared to classical hypothesis testing and density-based inference.

To summarize, our contributions are as follows:

- We introduce Membership Circuits[1], a tractable probabilistic model that assesses whether observations stem from a learned, multivariate distribution by computing exact p-values in a single bottom-up pass.
- We prove that MCs successfully bound the type I error, as they build on the concepts of p-variables and merging functions, resulting in valid, multivariate p-values.
- We evaluate the distributions of p-values as well as the type I error and power of the test empirically on synthetic and real data on the task of OOD detection.
- We demonstrate that MCs are able to detect if they failed to learn an accurate representation of the data.

2. Background

In this section, we establish the formal notation and theoretical foundations for hypothesis testing and PCs.

[1]Our code is available here.

Notation. We denote random variables as uppercase letters (X, Y) and their realizations as lowercase letters (x, y). Multivariate extensions are indicated in boldface, e.g., $\boldsymbol{X}$ and $\boldsymbol{x}$. A probability distribution is denoted P_X, with its associated probability density function (PDF) and cumulative distribution function (CDF) denoted as p_X and F_X, respectively. For brevity, densities refer to both probability density and mass functions, if applicable. Nodes in PCs are represented by $\oplus$ (sum), $\otimes$ (product), and $\oslash$ (leaf). A generic node is denoted as $\odot$.

Proofs. Proofs can be found in appendix A.

2.1. Hypothesis Testing

The goal of hypothesis testing is to determine if collected data contains sufficient evidence to contradict a well-defined question about some property of the data, the *null hypothesis* H_0 (Lehmann & Romano, 2005). We consider tests that bound the type I error rate, which is the probability of falsely rejecting a true null hypothesis (also referred as *false positive rate* in the following text). The bound on the type I error, the *test niveau* $\alpha \in [0, 1]$, is defined by the applicant of the test. The choice of α depends on the problem at hand and should consider balancing the acceptable false positive and true positive rates, as both are usually correlated.

After defining the null hypothesis H_0 and choosing the test niveau α, we compute a test statistic $t = t(x)$ from data x. By estimating the test distribution T, we can evaluate the probability that an event at least as extreme as the observed one occurs, given that H_0 is true. This probability is called the p-value $\hat{p} = \hat{p}(x)$ and defined w.r.t. the side of the test:

$$\hat{p}(x) := \begin{cases} P_T(T \leq t | H_0) =: l & \text{left-sided,} \\ P_T(T \geq t | H_0) =: r & \text{right-sided,} \\ 2 \cdot \min(l, r) & \text{two-sided.} \end{cases} \tag{2}$$

The null hypothesis is rejected if $\hat{p} \leq \alpha$.

The Multiple Testing Problem. When we wish to combine k many p-values into one representing the overall evidence, we are faced with the multiple testing problem (Bender & Lange, 2001). Choosing the minimum p-value among a set of tests would inflate the rate of falsely rejecting the null hypothesis beyond the test niveau α and violate the assumptions of our testing framework. Specifically, the resulting error rate can be computed as $1-(1-\alpha)^k$. A simple countermeasure is the Bonferroni correction, which adjusts the resulting p-value for the number of tests conducted: $\hat{p}_{\text{Bonf}} = k \cdot \min(\hat{\boldsymbol{p}})$, where $\hat{\boldsymbol{p}} \in [0,1]^k$ are the p-values of the respective tests (Bland & Altman, 1995). The Bonferroni correction yields valid test combination procedures without making any assumptions about the dependence structure of the tests. However, it is also known to be overly conservative, i.e., placing a too strict bound on the test niveau at the cost of sacrificing power (*true positive rate*). Vovk & Wang (2020) and Winkler et al. (2016) provide overviews of different p-value-merging functions and their assumptions.

2.2. Probabilistic Circuits

A Probabilistic Circuit $\mathcal{C}$ (Choi et al., 2020) is a computational graph encoding a joint probability distribution $p_{\mathcal{C}}(\boldsymbol{X})$ over random variables $\boldsymbol{X}$. A PC consists of 3 types of nodes:

1. Leaf nodes ($\oslash$) encode univariate parametric distributions (e.g., Gaussian or Poisson).
2. Sum nodes ($\oplus$) compute convex combinations of their k children, encoding mixture distributions: $p_{\oplus}(\boldsymbol{x}) = \sum_i w_i p_{\odot_i}(\boldsymbol{x})$, where $w_i \sim \Delta^{k-1}$ (and Δ^{k-1} is the standard (k-1)-simplex $\Delta^{k-1} = \{\mathbf{x} \in \mathbb{R}^k \mid x_i \geq 0, \sum_{i\in[1,..,k]} x_i = 1\}$)
3. Product nodes ($\otimes$) encode product distributions assuming statistical independence, $p_{\otimes}(\boldsymbol{x}) = \prod_i p_{\odot_i}(\boldsymbol{x})$.

PCs are typically evaluated bottom-up: the leaf nodes are evaluated first and propagate their values upwards until a root is reached, where inner nodes aggregate their children's values according to the procedures specified above. Each node $\odot$ is defined over a scope $\psi(\odot) \subseteq \boldsymbol{X}$. The scope of a leaf node is its associated random variable; the scope of an inner node is the union of the scopes of its children. To ensure tractability in PCs, i.e., exact and linear-time inference, two structural constraints are typically imposed: *smoothness*, i.e., for every sum node, all children must have identical scope, and *decomposability*, i.e., for every product node, all children must have pairwise disjoint scopes. In smooth and decomposable PCs, joint and marginal densities $p_{\mathcal{C}}(\boldsymbol{X'} = \boldsymbol{x'}), \boldsymbol{X'} \subseteq \boldsymbol{X}$ can be computed exactly and tractably in a single bottom-up pass in $\mathcal{O}(|\mathcal{C}|)$ time, where $|\mathcal{C}|$ is the number of edges in the circuit $\mathcal{C}$.

3. Membership Circuits

Our goal is to answer the hypothesis that a given multivariate sample originates from a learned joint distribution $P_{\mathcal{C}}$ in a tractable way with probabilistic guarantees, i.e.,

$$H_0 : \boldsymbol{x} \sim P_{\mathcal{C}}. \tag{3}$$

The alternative hypothesis is then

$$H_1 : \boldsymbol{x} \nsim P_{\mathcal{C}}. \tag{4}$$

Learning $P_{\mathcal{C}} \approx P_{\boldsymbol{X}}$ offers two advantages over traditional methods: first, the membership test is less limited to specific representations of the data. In particular, it allows mixing different distributional families of a large number of RVs rather than being constrained to a single variable of interest. Second, the test harnesses the expressivity of deep learning to construct test distributions that approximate the data well. PCs enable capturing nuanced relationships, such as context-specific independencies (Zhang & Poole, 1999), resulting in more faithful approximations than global parametric models, which may improve the power of the test compared to traditional methods.

Since we want to assess the relationship of an observation $\boldsymbol{x}$ to our learned distribution $P_{\mathcal{C}}$, we define it as our test distribution T:

$$t = \boldsymbol{x} \quad \text{and} \quad T = P_{\mathcal{C}} \approx P_{\boldsymbol{X}}. \tag{5}$$

Traditionally, this would imply that we construct either Union-over-Intersection (UIT) or Intersection-over-Union (IUT) tests, depending on requiring all or just any of the random variables being more extreme than the test statistic t. However, utilizing the hierarchical structure of MCs allows us to construct a more fine-grained test procedure, where we can ask how likely an observation is in the joint distribution and to interpolate between the ”extreme” IUT and UIT scenarios. Given a learned PC $P_{\mathcal{C}}$, we convert the learned model into an MC $M_{\mathcal{C}}$ by replacing the functions that compute the nodes in $P_{\mathcal{C}}$ so that we compute valid and exact p-values: First, univariate p-values are computed in the leaves. Then, merging functions combine p-values of mixture distributions or product distributions at inner nodes. Since $P_{\mathcal{C}}$ and $M_{\mathcal{C}}$ share the same structural properties, the computation of p-values is tractable. Next, we lay the theoretical foundations for MCs, introduce MCs formally and show that they compute sound p-values.

3.1. *p*-variables and Merging Functions

To ensure that the membership test is valid, i.e., respects the bound on the type I error, we build on Vovk & Wang (2020)'s concepts of p-*variables* and *merging functions*.

Definition 3.1 (p-variable)**.** A random variable X is a p-*variable* if it satisfies

$$P(X \leq \epsilon) \leq \epsilon, \quad \forall \epsilon \in (0,1). \tag{6}$$

	Sum Nodes ($\oplus$)	Product Nodes ($\otimes$) $\hat{p}_i$ indep.	$\hat{p}_i$ dep.
Aggregate	Ar. Mean (Eq. 9)	Fisher (Eq. 12)	H. Mean (Eq. 15)
Select	$\max \mathcal{L}$ (Eq. 11)	Tippett (Eq. 13)	Bonferroni (Eq. 14)

Table 2. Membership Circuits can choose among different merging functions suited for aggregating or selecting tests.

In other words, a p-variable is a random variable whose values are (possibly conservative) p-values. Note that this definition is equivalent to X first-order stochastically dominating the standard uniform distribution $\mathcal{U}(0, 1)$. To formalize the combination of multiple random variables into a single p-variable, we introduce a relaxed version of the definition of merging functions by Vovk & Wang (2020).

Definition 3.2 (Merging function)**.** An increasing Borel function $f : [0, 1]^k \to [0, 1]$ is a *merging function* of random variables $X_1, \dots, X_k$, if the following holds:

$$P(f(X_1, \dots, X_k) \leq \epsilon) \leq \epsilon, \quad \forall \epsilon \in (0, 1). \tag{7}$$

Our relaxation, requiring that only one out of the k random variables is a p-variable rather than all of them, will be necessary for computing p-values of sum nodes.

3.2. Computing *p*-values

The query $\hat{p}_{\mathcal{C}}(\boldsymbol{x})$ is evaluated in a single bottom-up pass. We define now the functions for computing p-values at leaf, sum, and product nodes, respectively.

Leaf nodes. We compute the p-values of the univariate distributions in the leaves by evaluating observations in the respective CDFs w.r.t. the chosen test side. For a leaf $\oslash$ and a realization x such that $\psi(\oslash) = X$, we define

$$\hat{p}_{\oslash}(x) \coloneqq \begin{cases} F_{\oslash}(x) & =: l \quad \text{left-sided}, \\ 1 - F_{\oslash}(x) & =: r \quad \text{right-sided}, \\ 2 \cdot \min(l, r) & \quad\quad \text{two-sided}. \end{cases} \tag{8}$$

We show that $\hat{p}_{\oslash}$ computes valid p-values for a random variable X via a p-variable argument.

Proposition 3.3. *Let $\oslash$ be a leaf node encoding a univariate probability distribution associated with a random variable X, an existing CDF $F_{\oslash}$, and $\hat{p}_{\oslash} : \mathbb{R} \to [0, 1]$ defined as above. Then the random variable $\hat{p}_{\oslash}(X)$ is a p-variable.*

Sum nodes. At sum nodes, the set of children's p-values needs to be merged into a p-value of the encoded mixture distribution. Similar to computing densities in sum nodes, it would appear natural to compute the arithmetic mean of the densities of its children, weighted by the sum weights $\boldsymbol{w} \sim \Delta^{k-1}$. However, p-values cannot be simply multiplied: The multiple testing problem (see Section 2.1) demonstrates that doing so would inflate the number of false rejections and invalidate the test. Vovk & Wang (2020) have shown which merging functions output p-variables if all inputs are p-variables. But this assumption does not hold in mixture distributions, where an observation $x \sim P$ is usually distributed w.r.t. only one of the mixture components. Hence, only one child of the sum node is a p-variable. Since we do not know which one, we have to introduce a correction term that ensures that the random variable encoded at the sum node is again a p-variable. Specifically, for a sum node $\oplus$ with weights $\boldsymbol{w}$, number of children k, and their respective p-values $\hat{\boldsymbol{p}} = \{\hat{p}_i\}_{i=1}^k$:

$$\hat{p}_{\oplus,\mathrm{AM}}(\boldsymbol{x}) = \frac{1}{\min \boldsymbol{w}} \cdot \sum\nolimits_{i=1}^{k} w_i \hat{p}_i. \tag{9}$$

Note that the arithmetic mean aggregates evidence across all children into a single statistic. In scenarios where we wish to test H_0 against the best-fitting latent explanation, we can adopt a maximum-likelihood-based selection rule that identifies the respective sub-circuits by evaluating the density function:

$$w_i^* = \begin{cases} 1 & \text{if } i = \operatorname{argmax}_{i \in \{1,\dots,k\}} w_i p_i(\boldsymbol{x}), \\ 0 & \text{else}. \end{cases} \tag{10}$$

This selection procedure then yields

$$\hat{p}_{\oplus,\max \mathcal{L}}(\boldsymbol{x}) = \sum\nolimits_{i=1}^{k} w_i^* \hat{p}_i = \hat{p}_{i^*} \tag{11}$$

with $i^* = \operatorname{argmax} w_i^*$. The selection method is particularly powerful to reject instances that are out-of-distribution w.r.t. the cluster that most likely could have produced it.

Proposition 3.4. *Let $\oplus$ be a sum node with k children and weights $\boldsymbol{w} = \{w_i\}_{i=1}^k$ s.t. $\boldsymbol{w} \in \Delta^{k-1}$ encoding a mixture distribution over random variables $\boldsymbol{X}$. Let one random variable $X_j, 1 \leq j \leq k$, be associated with a p-variable, and $\hat{\boldsymbol{p}} = \{\hat{p}_i\}_{i=1}^k$ be the values of the sum node's children. Then $\hat{p}_{\oplus,AM}$ and $\hat{p}_{\oplus,\max \mathcal{L}}$ are merging functions.*

Product nodes. At product nodes, the set of children's p-values has to be merged into p-values of the encoded product distributions. Multiplying p-values is not applicable, since it would inflate the type I error rate and violate the respective bound (compare right side of fig. 1).

Fisher (1932) showed that aggregating p-values of independent tests follows a χ^2 distribution with $2k$ degrees of freedom. If k sources of evidences $\hat{\boldsymbol{p}} = \{\hat{p}_i\}_{i=1}^k$ are to be aggregated at a product node $\otimes$, we compute

$$\hat{p}_{\otimes,\mathrm{Fisher}}(\boldsymbol{x}) = 1 - \chi^2_{\mathrm{CDF}} \left(-2 \sum\nolimits_{i=1}^{k} \ln \hat{p}_i;\ 2k \right) \tag{12}$$

Similar to the maximum likelihood selection method in Eq. 11, it may be preferable to not aggregate tests but to choose the most significant among them. To this end, the method of Tippett (1931) corrects the minimum p-value for the number of tests conducted:

$$\hat{p}_{\otimes,\text{Tippett}}(\boldsymbol{x}) = 1 - (1 - \min(\hat{\boldsymbol{p}}))^k. \tag{13}$$

Assuming statistical independence, both procedures are merging functions.

Proposition 3.5. *Let $\otimes$ be a product node with k pairwise statistically independent children encoding a product distribution over random variables $\boldsymbol{X}$. Let $\hat{\boldsymbol{p}} = \{\hat{p}_i\}_{i=1}^k$ be the p-values of the product node's children. Then $\hat{p}_{\otimes,\text{Fisher}}$ and $\hat{p}_{\otimes,\text{Tippett}}$ are merging functions.*

While product nodes formally encode statistical independence, this assumption often holds only approximately in practice. Merging functions that do not assume independent p-values may then compensate if the true underlying dependence structure is either unknown or known to be false. Similar to Tippett's method (Tippett, 1931), the Bonferroni correction (Bland & Altman, 1995) selects the most significant test and corrects it for the number of tests conducted without assuming independence among the partial tests:

$$\hat{p}_{\otimes,\text{Bonferroni}}(\boldsymbol{x}) = k \cdot \min(\hat{\boldsymbol{p}}). \tag{14}$$

To aggregate dependent tests, Wilson (2019) proposed the harmonic mean p-value. According to the authors, the test is more powerful than the Bonferroni correction while controlling the false positive rate. However, Goeman et al. (2019) demonstrated that the harmonic mean *per se* does not result in a valid test procedure. By introducing a correction term, Vovk & Wang (2020) showed that the corrected harmonic mean is a merging function again.

$$\hat{p}_{\otimes,\text{HM}}(\boldsymbol{x}) = \frac{a_k \cdot k}{\sum_{i=1}^k \hat{p}_i^{-1}} \tag{15}$$

with a_k being the correction factor depending on the number of tests to be combined:

$$a_k = \begin{cases} e \cdot \ln(2) & k = 2, \\ \frac{(y_k + k)^2}{(y_k + 1) \cdot k} & k > 2. \end{cases} \tag{16}$$

where y_k is the unique solution to the equation

$$y^2 = k \cdot ((y+1) \cdot \ln(y+1) - y), \quad y \in (0, \infty). \tag{17}$$

Proposition 3.6. *Let $\otimes$ be a product node with k children encoding a product distribution over random variables $\boldsymbol{X}$. Let $\hat{\boldsymbol{p}} = \{\hat{p}_i\}_{i=1}^k$ be the p-values of the product node's children. Then $\hat{p}_{\otimes,\text{Bonferroni}}$ and $\hat{p}_{\otimes,\text{HM}}$ are merging functions.*

3.3. MCs compute sound p-values

While the previous section has shown that nodes in MCs compute sound p-values w.r.t. their assumptions, we now establish that these assumptions hold in Membership Circuits and the root nodes encodes a p-variable. To this end, we utilize the induced tree representation. An induced tree τ is a subgraph of $P_\mathcal{C}$ that traverses the circuit starting from the root node and contains a) the root node, b) all children of product nodes, and c) exactly one child of sum nodes. The set of all induced trees of a circuit is $\mathcal{T}$.

By definition, a sample that originates from the circuit's encoded distribution has to be encoded by one induced tree. Since each node in this induced tree encodes exactly the distributions that $\boldsymbol{x}$ stems from, the respective nodes encode p-variables. So, given that $\boldsymbol{x} \sim P_\mathcal{C}$, then there exists a $\tau^* \in \mathcal{T}$ s.t. $\boldsymbol{x}$ is distributed w.r.t. all nodes in τ^* and hence all nodes in τ^* are p-variables. Since our merging functions at sum nodes only assume one of the children to encode a p-variable to conserve the p-variable property, all sum nodes in τ^* are p-variables. Therefore, if the root node is a sum node, then the it encodes a p-variable. If the root node is a product node, then all its children, either sum nodes or leaf nodes, encode p-variables. Hence, the root node of an MC encodes a p-variable and computes sound p-values, which establishes the validity of the proposed test. [2]

3.4. Detection of Model Mismatches

Membership Circuits provide a built-in diagnostic tool to identify model mismatches. If the MC is an accurate representation of the target distribution, the merging functions ensure that the membership test is sound, i.e. $\text{FPR} \leq \alpha$. If, however, the test yields a false positive rate significantly higher than α when evaluated on the training data, this result serves as a diagnostic signal that the circuit has failed to faithfully approximate the respective distribution. Specifically, as the test utilizes the learned distribution, a signal indicates either high bias of the learned model w.r.t. the target distribution (underfitting), or that the model was unable to adequately capture the variance of the data. Note that, in the sense of statistical testing methodology, a successful test does not necessarily indicate a good architecture. However, this diagnostic tool is a computationally cheap "sanity check", utilizing the tractability of the test, and may anticipate otherwise expensive model checking techniques.

4. Empirical Evaluation

To investigate the validity of the membership test, its potential on the task of out-of-distribution detection in machine learning settings, as well as its ability to identify model

[2]A formalized version is in development.

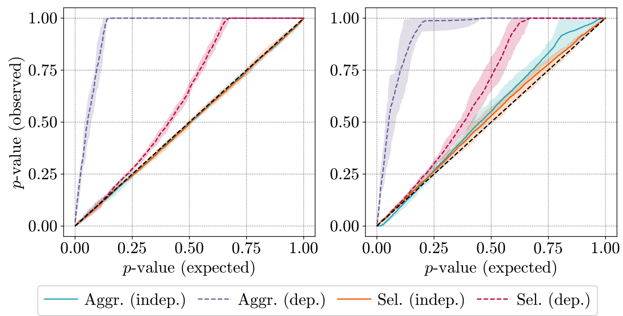

Figure 2. **MCs guarantee to compute sound p-values.** Quantile-quantile plots of merging functions. If observed p-values are at least as high as the respective expected p-values, the test is valid. Mean and min./max. values across 5 repetitions. ***Left:*** MC learned from uncorrelated data. ***Right:*** MC learned from correlated data.

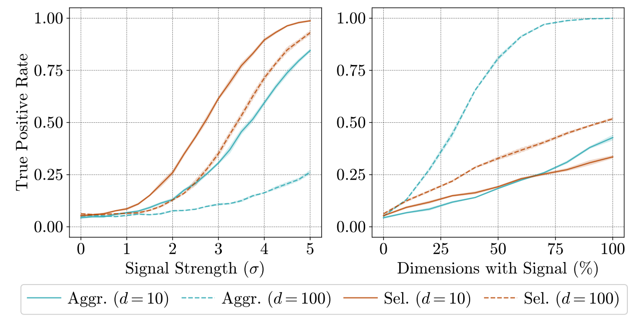

Figure 3. **MCs reliably identify signals in high-dimensional data.** Mean power ($\pm$ st. dev.) across 5 repetitions on OOD detection. ***Left:*** A signal of strength σ is injected into a one dimension of the test data. ***Right:*** Signals of strength 1 are injected into a fraction of all dimensions of the test data.

mismatches, we used the following merging functions:

- Aggregation (indep.): $\oplus$: AM, $\otimes$: Fisher
- Aggregation (dep.): $\oplus$: AM, $\otimes$: HM
- Selection (indep.): $\oplus$: $\max \mathcal{L}$, $\otimes$: Tippett
- Selection (dep.): $\oplus$: $\max \mathcal{L}$, $\otimes$: Bonferroni

Using them, we investigated the following questions:

(Q1) Are the p-values computed by MCs empirically sound?

(Q2) How well perform different merging functions in detecting differences in multivariate data?

(Q3) Can MCs empirically guarantee the user-defined bound on the FPR?

(Q4) How do MCs perform when data is missing?

(Q5) Does a violation of the type I error bound indicate that the model failed to learn an accurate representation?

(Q1) Soundness of p-values. We investigated the distributions of p-values evaluated in circuits learned from data that consists of a) globally independent RVs and b) strongly correlated RVs. We generated train and test data consisting of $n = 1000$ samples each from a bimodal 10-dimensional Gaussian distribution. For a), we use $\mu_i \in \{-1, 1\}$ per dimension and $\Sigma = \boldsymbol{I}$. For b), we add pairwise covariances of 0.5. We used LearnMSPN (Molina et al., 2018) with a minimum of 50 instances per leaf and an RDC threshold of 0.3 (Lopez-Paz et al., 2013) to learn MCs with Gaussian leaves. We conducted two-sided tests in both settings.

The resulting quantile-quantile plots are presented in Fig. 2. The black dashed line represents the theoretical optimum producing perfectly uniformly distributed p-values. If the independence assumption in the data and the circuits holds (*left*), all merging functions are either equal to or stochastically dominate the standard uniform distribution (curves stay on or over the black dashed line). This confirms their ability to preserve the p-variable property, resulting in often conservative but valid tests. In the correlated case (*right*), the independent aggregation violates the bound in the interval $[0, 0.15]$. The selection methods are surprisingly robust.

(Q2) Power Analysis. Since the sample size of the proposed membership test is fixed to 1, we investigated its power w.r.t. a) single strong signals (one dimension shifted by σ), and b) multiple weak signals (a fraction of dimensions shifted by $\sigma = 1$). In both cases, we sampled training data of size $n = 1000$ from $d \in \{10, 100\}$ independent bimodal Gaussian distributions as a transparent, easy-to-follow setup. We used LearnMSPN with 50 observations per leaf and an RDC threshold of 0.3 to learn MCs with Gaussian leaves. The test data is then sampled from the following modified multivariate Gaussian distributions: a) We increased the mean of 1 out of the d dimensions by signal strength $\sigma \in \{0.25, 0.5, \ldots, 5.0\}$. b) We increased the mean of $r \in \{10\%, 20\%, \ldots, 100\%\}$ of the d dimensions by signal strength 1. The power (true positive rate) is the proportion of the test samples correctly rejecting H_0.

Fig. 3 shows the behavior of the merging functions assuming independence in two-sided testing. For single strong signals (*left*), the selection method for $d = 10$ correctly rejects the null hypothesis in 99% of the cases of signals of strength $\sigma = 5$ and around 60% of the cases when a signal of strength $\sigma = 3$ is injected. While 3 standard deviations are a usual outlier threshold in univariate cases, the test demonstrates strong results considering the number of dimensions the test is carried out in. With an increasing number of dimensions, the test expectantly performs worse. However, the selection method demonstrates its ability to find the "needle in the haystack" by identifying anomalies where only 1% of the variables is corrupted, provided the signal is strong.

Dataset	MCs @ $\alpha = 5\%$ FPR (↓, %)		Power (↑, %)
aloi	0.96 - 1.13	✓	3 - 5
annthyroid	8.72 - 9.83	✗	9 - 18
b-cancer	5.60 - 8.40	✗	40
kdd99	0.11 - 0.91	✓	50 - 95
letter	1.60 - 3.27	✓	30 - 45
pen-global	1.81 - 3.06	✓	13 - 16
pen-local	2.58 - 4.81	✓	60 - 90
satellite	4.98 - 6.23	✓	32 - 37
shuttle	0 - 0.07	✓	1 - 2
speech	10.3 - 32.9	✗	11 - 26

Table 3. **MCs provide probabilistic guarantees.** They either bound the false positive rate (✓) or, if they are unable to (i.e., $\text{FPR} > \alpha$), know when the underlying PC is an inaccurate representation of the data (✗).

Dataset	AuROC (↑, %) LOF	MC (50% MV)	MC (80% MV)
aloi	N/A	52.09 ± 1.60	50.80 ± 2.07
annthyroid	N/A	58.06 ± 13.9	61.39 ± 16.1
b-cancer	N/A	90.40 ± 16.7	88.29 ± 18.1
kdd99	N/A	89.88 ± 7.08	89.33 ± 9.73
letter	N/A	68.05 ± 4.80	59.25 ± 6.92
pen-global	N/A	84.47 ± 6.43	74.57 ± 8.31
pen-local	N/A	84.04 ± 7.25	73.38 ± 15.3
satellite	N/A	94.39 ± 0.75	93.47 ± 1.58
shuttle	N/A	87.17 ± 15.1	91.40 ± 12.9
speech	N/A	50.23 ± 1.05	49.11 ± 1.13

Table 4. **MCs can handle missing data.** While other OOD detection methods like the LOF require complete observations, MCs allow inference with missing data. Depicted are mean ± std. dev. of AuROC across 5 repetitions.

For multiple weak signals (*right*), the aggregation method successfully combines anomalous evidence spread out across many dimensions. For $d = 100$, its power converges to 1. It is able to identify half of the out-of-distribution instances as such if 25% of the dimensions are affected and approximately 90% of instances if half of the dimensions are affected. Especially in scenarios where instances are *jointly* anomalous but not in the respective marginal distributions, we argue that the membership test can automate out-of-distribution detection in high-dimensional spaces with superhuman performance. This is particularly useful for systems that monitor many variables of interest to assess if any events requiring human intervention occur, e.g., control stations in power or purification plants.

(Q3) MCs provide error control. To evaluate the test's ability in real-world scenarios, we conducted experiments on 10 unsupervised outlier detection datasets (Goldstein & Uchida, 2016). The models are learned from the entire datasets without having access to the respective labels. Then, each instance is assigned a score regarding its anomaly. We conducted two-sided membership tests with independent aggregation. We compared MCs with densities computed in the same model (PC (Molina et al., 2018)) and a dedicated outlier detection method (LOF (Breunig et al., 2000)). All methods are evaluated w.r.t. the area under the receiver operating characteristic (AuROC) on sets of common choices of hyperparameters (see app. C for details). Also, we measure the FPR and the power of the test at the test level $\alpha = 5\%$.

Tab. 3 shows that MCs can bound the FPR w.r.t. the chosen test niveau even on real-world datasets (indicated by ✓). Although this bound only hold approximately because of the approximate nature of the model, the results demonstrate the reliability of our method. On some datasets, the test has low power. This may be attributed primarily to two reasons: First, the underlying circuit does not represent the data sufficiently well, which influences the quality of the test. Second, the correction factor of the arithmetic mean leads to overly conservative p-values, which can also be observed in Fig. 2 (right side). However, the observed FPR significantly exceeding α indicate poor model fit without extensive ad-hoc methods (cf. Fig 4).

(Q4) MCs can handle missing data. Traditional methods like LOF are not well-defined when encountering missing data, i.e., they require expensive processing steps like imputation. MCs, on the other hand, allow for efficient marginalization and hence for inference when data is only partially observed. To this end, we repeated the experiment of unsupervised outlier detection (Q3) with 50% and 80% of the observable variables missing. We removed random partitions in $\%$ repeated 10 times per dataset and model.

Tab. 4 shows the results of the membership test on partial data. For LOF, no results are reported, since the method is unable to perform inference on incomplete observations. For MC, the performance is often only affected slightly. At 80% missing variables, most models still report an average Area under the Receiver Operating Characteristic of around 75% or higher. In some cases, e.g. on the dataset "shuttle", the performance increases. While this may be attributed to removing noise and hence reducing the data to the signal determining OOD, the variance in the observed cases is noticeably high. The distributions of the respective results are right-skewed, where a few runs of the experiment lead to increased performance, but the average score decreases if the respective outliers are removed.

(Q5) MCs detect model mismatch. MCs guarantee to bound the FPR w.r.t. the test niveau α if the circuit represents the data accurately. If, however, the FPR exceeds α, this signal indicates that the learned architecture is not a faithful approximation of the data. To support this claim

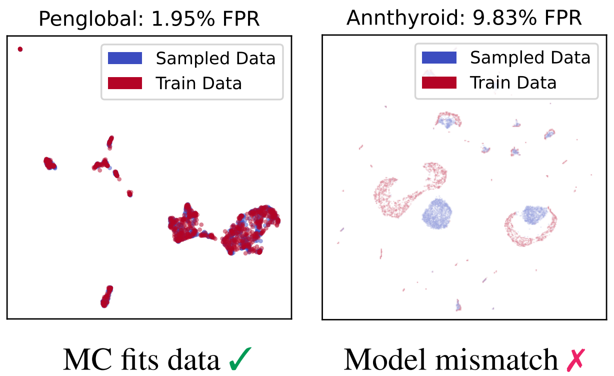

Figure 4. **Membership circuits know when the underlying architecture is wrong.** UMAP visualizations of original and generated data demonstrate that a violation of the FPR bound indicates poor model fit. ***Left:*** Samples from circuit explore the original distribution well. ***Right:*** Samples belong to a different distribution than the original dataset.

empirically, we selected models from the OOD detection experiments that violate the bound on the FPR, cf. Fig. 4. We then generated samples of the same size as the original training data the circuit was learned from. The original data and samples from the circuit are concatenated and visualized with UMAP (McInnes et al., 2018).

5. Related Work

The integration of statistical hypothesis testing with machine learning has led to two distinct approaches: using hypothesis tests to evaluate learned models and using DL to compute or improve test statistics. While a substantial amount of work has been dedicated to the former (Dietterich, 1998; Nadeau & Bengio, 1999; Bouckaert & Frank, 2004; Horel & Giesecke, 2020; Mandel & Barnett, 2024; Shi et al., 2024), we focus on the methods regarding the latter.

Kirchler et al. (2020) proposed to estimate the kernels for Gretton et al. (2012)'s Maximum Mean Discrepancy with neural networks. While their work allows for comparing two distributions, our test assesses the membership of single instances against a learned distribution. Moreover, neural tests usually require expensive permutation testing to estimate test distributions whereas our PC-based approach provides analytical p-values tractably.

Kulessa et al. (2021) have first compute p-values in PCs, but did not account for the their unique characteristics and lacked the formal construction of a hypothesis test with probabilistic guarantees. Our work can be seen as their theoretical completion by incorporating the theory of p-variables and merging functions developed by Vovk & Wang (2020). Broadrick et al. (2024) investigated PCs representing joint CDFs, while MCs follow a divide and conquer approach to merge p-values from univariate CDFs in the leaves.

Scott & Nowak (2005)'s approach to density level set estimation learns the decision region directly from data by optimizing for the minimum-volume region needed for a targeted test niveau. Our method doesn't optimize the learned model w.r.t. the rejection region, but is an inference routine that offers testing for membership in circuits that are learned to model the density of multivariate training data.

Although reliability is at the heart of probabilistic machine learning, multiple works have shown that deep generative models are overly confident. Ventola et al. (2023) introduced dropout-like mechanisms to estimate query-wise variances in RAT-SPNs (Peharz et al., 2020). Their method aims to improve their reliability by estimating model uncertainty; our method provides guarantees in models that represent the target distribution fairly accurately and is able to detect if the models fail to do so. The overconfidence statement is reinforced by Nalisnick et al. (2019), who developed a test assessing if a batch of samples belongs to the typical set of a deep generative model. Their approach is a two-sample test between data and a learned distribution, while our method focuses on identifying if single instances belong to a model.

6. Conclusions and Future Work

We presented Membership Circuits, a novel model that assesses the membership hypothesis $H_0 : \boldsymbol{x} \sim P_\mathcal{C}$ in multivariate distributions encoded by PCs. The mathematically rigorous inference routine computes multivariate p-values tractably by leveraging structural properties. Merging functions ensure the test is sound and the user-defined type I error bound is respected. Our experiments provided insights into the distributions of p-values, the power of the test, and its ability for reliable OOD detection on real-world datasets. If the test fails to provide its guarantees, it may be used as a diagnostic signal assessing the quality of the underlying model. The demonstrated properties make MCs a strong candidate for safety-critical monitoring systems and facilitate a general yet principled approach to quantifying whether data stems from a known, complex distribution.

The applicability and statistical power of our proposed membership test are subject to four primary technical constraints. First, the framework requires the availability of tractable Cumulative Distribution Functions (CDFs) at the leaf nodes of the MC to ensure exact computation. Second, the test's reliability is inherently coupled with the model's representative capacity; any significant approximation error in the learned distribution P may lead to miscalibrated inference. Third, our test's rejection region is induced implicitly by the learned circuit and choice of merging functions, and its geometric properties remain uncharacterized. Lastly, we observe potential loss of power in high-dimensional settings, since accounting for dependencies among variables may lead to overly conservative p-values, requiring increasingly distinct signals for successful detection.

Future work may develop more expressive architectures or tighter bounds on dependency-aware merging functions. Composite merging functions or e-values (Ramdas & Wang, 2025) may reduce the conservativeness of deep MCs. Measuring the divergence between expected and observed p-values could serve as model selection metric or regularizer, while learned test distributions may further bridge statistical rigor with the expressiveness of deep learning.

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Software and Data

Our code and instructions for acquiring data used in experiments can be found at `https://github.com/bewit/membership_circuits`.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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A. Proofs

We present formal proofs for the propositions and corollaries regarding the functions introduced by the membership test. To this end, we reintroduce the necessary functions computing p-values at leaf, sum, and product nodes.

Leaf nodes compute sound p-values

Proposition A.1. Let $\otimes$ be a leaf node encoding a univariate probability distribution associated with random variable X and existing CDF $F_{\otimes}$, and $\hat{p}_{\otimes} : \mathbb{R} \to [0, 1]$ be defined as

$$\hat{p}_{\otimes}(x) := \begin{cases} F_{\otimes}(x) & =: l \quad \text{left-sided,} \\ 1 - F_{\otimes}(x) & =: r \quad \text{right-sided,} \\ 2 \cdot \min(l, r) & \quad\quad\; \text{two-sided.} \end{cases}$$

Then the random variable $\hat{p}_{\otimes}(X)$ is a p-variable.

Proof. We consider three cases: left-sided, right-sided, and two-sided p-values. In all cases, we assume that the CDF $F_{\otimes}(X)$ is an accurate approximation to the true CDF of X, i.e. the approximation error can be bound by an arbitrary small constant.

Left-sided case: We show that $P(F_{\otimes}(X) \leq \alpha) \leq \alpha$ by applying the probability integral transform. Let $F_{\otimes}^{-1}(\alpha) = \inf\{x \in \mathbb{R} : F_{\otimes}(x) \geq \alpha\}$ be the generalized inverse quantile function. Then $F_{\otimes}(x) \leq \alpha \Leftrightarrow x \leq F_{\otimes}^{-1}(\alpha), \forall x \in X$. For a continuous RV X, we obtain

$$P(F_{\otimes}(X) \leq \alpha) = P(X \leq F_{\otimes}^{-1}(\alpha)) = F_{\otimes}(F_{\otimes}^{-1}(\alpha)) = \alpha$$

$F_{\otimes}(X)$ is hence uniformly distributed, $F_{\otimes}(X) \sim \mathcal{U}(0, 1)$. If X is discrete, then the last equality turns into a less-or-equal, because the CDF of a discrete RV is only right-continuous and non-decreasing (rather than continuous and strictly increasing), leading to $F_{\otimes}(X)$ stochastically dominating $\mathcal{U}(0, 1)$.
In both cases, $F_{\otimes}(X)$ is a p-variable.

Right-sided case: We show that $1 - F_{\otimes}(X) \leq \alpha$ by a symmetry argument. If X is continuous, then $F_{\otimes}(X) \sim \mathcal{U}(0, 1)$ as well as $1 - F_{\otimes}(X) \sim \mathcal{U}(0, 1)$. In the discrete case, let $G_{\otimes}(X) = 1 - F_{\otimes}(X) = P(X \geq x)$ be the complementary function to the respective CDF with existing inverse $G_{\otimes}^{-1}$. $G_{\otimes}$ is right-continuous and non-increasing. Following the argument for the left-sided case, we obtain:

$$P(1 - F_{\otimes}(X) \leq \alpha) = P(G_{\otimes}(X) \leq \alpha) = P(X \leq G_{\otimes}^{-1}(\alpha)) = G_{\otimes}(G_{\otimes}^{-1}(\alpha)) \leq \alpha$$

Hence, $1 - F_{\otimes}(X)$ is a p-variable in both cases.

Two-sided case: We show that $2 \cdot \min(F_{\otimes}(X), 1 - F_{\otimes}(X)) \leq \alpha$ by a merging function argument. We combine two p-variables $F_{\otimes}(X)$ and $1 - F_{\otimes}(X)$ by correcting them by a factor of 2. This is an instantiation of the Bonferroni correction, which is known to be a merging function, i.e., its output is a p-variable whenever its inputs are p-variables (compare (Vovk & Wang, 2020), Proposition 8). We have shown before that both $F_{\otimes}(X)$ and $1 - F_{\otimes}(X)$ are p-variables.
Hence, $2 \min(F_{\otimes}(X), 1 - F_{\otimes}(X))$ is a p-variable. $\square$

Sum nodes compute sound p-values

At sum nodes, we either compute $\hat{p}_{\oplus,\mathrm{AM}}$ to aggregate the tests of the sum node's children or $\hat{p}_{\oplus,\max\mathcal{L}}$ to select the most significant among them.

$$\hat{p}_{\oplus,\mathrm{AM}}(\boldsymbol{x}) = \frac{1}{\min(\boldsymbol{w})} \cdot \sum\nolimits_{i=1}^{k} w_i \hat{p}_i$$

$$w_i^* = \begin{cases} 1 & \text{if } i = \mathrm{argmax}_{i \in \{1,\dots,k\}} w_i p_i(\boldsymbol{x}), \\ 0 & \text{else.} \end{cases}$$

$$\hat{p}_{\oplus,\max\mathcal{L}}(\boldsymbol{x}) = 1 \cdot \sum\nolimits_{i=1}^{k} w_i^* \hat{p}_i = \hat{p}_{i^*}, \quad i^* = \mathrm{argmax}\, w_i^*$$

Proposition A.2. Let $\oplus$ be a sum node with k children and weights $\boldsymbol{w} = \{w_i\}_{i=1}^{k}$ s.t. $\boldsymbol{w} \in \Delta^{k-1}$ encoding a mixture distribution over random variables $\boldsymbol{X}$. Let one random variable $\boldsymbol{X}_j, 1 \leq j \leq k$, be associated with a p-variable, and $\hat{\boldsymbol{p}} = \{\hat{p}_i\}_{i=1}^{k}$ be the values of the sum node's children. Then $\hat{p}_{\oplus,\mathrm{AM}}$ and $\hat{p}_{\oplus,\max\mathcal{L}}$ are merging functions.

Proof. We show that $\hat{p}_{\oplus,\mathrm{AM}}$ by bounding it w.r.t. the p-variable component X_j. By definition of a mixture distribution, if an observation $\boldsymbol{x} \sim P(\boldsymbol{X})$, then it is distributed w.r.t. one of the mixture components, $\boldsymbol{x} \sim P(\boldsymbol{X}_j)$. This $\boldsymbol{X}_j$ is then associated with a p-variable $\hat{p}(\boldsymbol{X}_j)$.

Since all $\hat{p}(\boldsymbol{X}_i)$ and w_i are bounded in $[0, 1]$, it holds that $w_j\hat{p}(\boldsymbol{X}_j) \leq \sum_{i=1}^{k} w_i\hat{p}(\boldsymbol{X}_i)$, i.e., the weighted p-variable is stochastically dominated by the weighed arithmetic mean of all children of the sum node. Now, we don't know which child encodes the p-variable $\hat{p}(\boldsymbol{X}_j)$. The worst case is that $\hat{p}(\boldsymbol{X}_j)$ is associated with the smallest weight of the sum node, meaning that it might be the least favorable distribution having the smallest impact that we still need to account for. This implies the lower bound $\min_{i=1\ldots k}(w_i)\,\hat{p}(\boldsymbol{X}_j) \leq w_j\hat{p}(\boldsymbol{X}_j)$.

By combining both bounds and rearranging the inequality for $\hat{p}(\boldsymbol{X}_j)$, we obtain the merging function $\hat{p}_{\oplus,\mathrm{AM}}$:

$$\begin{aligned}
&\min_{i=1\ldots k}(w_i)\,\hat{p}(\boldsymbol{X}_j) \leq w_j\hat{p}(\boldsymbol{X}_j) \leq \sum_{i=1}^{k} w_i\hat{p}(\boldsymbol{X}_i)\\
\iff &\min_{i=1\ldots k}(w_i)\,\hat{p}(\boldsymbol{X}_j) \leq \sum_{i=1}^{k} w_i\hat{p}(\boldsymbol{X}_i)\\
\iff &\hat{p}(\boldsymbol{X}_j) \leq \frac{1}{\min_{i=1\ldots k}(w_i)} \sum_{i=1}^{k} w_i\hat{p}(\boldsymbol{X}_i) =: \hat{p}_{\oplus,\mathrm{AM}}(\boldsymbol{X}).
\end{aligned}$$

Since $\hat{p}(\boldsymbol{X}_j) \leq \hat{p}_{\oplus,\mathrm{AM}}(\boldsymbol{X})$, i.e., the weighted arithmetic mean merging function stochastically dominates the p-variable component, the probability of the merging function being bounded by α is bounded by the p-variable component being bounded by α:

$$\mathbb{P}(\hat{p}_{\oplus,\mathrm{AM}}(\boldsymbol{X}) \leq \alpha) \leq \mathbb{P}(\hat{p}(\boldsymbol{X}_j) \leq \alpha) \leq \alpha.$$

Hence, $\hat{p}_{\oplus,\mathrm{AM}}$ is a merging function.

□

Product nodes compute sound p-values

At product nodes, we can aggregate or select tests w.r.t. their underlying dependence structure.

If the tests to be combined are statistically independent, then we choose $\hat{p}_{\otimes,\mathrm{Fisher}}$ for aggregation or $\hat{p}_{\otimes,\mathrm{Tippett}}$ for selection. If they are not, we choose $\hat{p}_{\otimes,\mathrm{HM}}$ for aggregation or $\hat{p}_{\otimes,\mathrm{Bonferroni}}$ for selection.

$$\hat{p}_{\otimes,\mathrm{Fisher}}(\boldsymbol{x}) = 1 - \chi^2_{\mathrm{CDF}}\left(-2 \cdot \sum\nolimits_{i=1}^{k} \ln(\hat{p}_i);\ 2k\right)$$

$$\hat{p}_{\otimes,\mathrm{Tippett}}(\boldsymbol{x}) = 1 - (1 - \min(\hat{\boldsymbol{p}}))^k$$

Proposition A.3. Let $\otimes$ be a product node with k pairwise statistically independent children encoding a product distribution over random variables $\boldsymbol{X}$. Let $\hat{\boldsymbol{p}} = \{\hat{p}_i\}_{i=1}^{k}$ be the p-values of the product node's children. Then $\hat{p}_{\otimes,\mathrm{Fisher}}$ and $\hat{p}_{\otimes,\mathrm{Tippett}}$ are merging functions.

Proof. We consider Tippett's method first. Let $X_1, \ldots, X_k$ be p-variables and $M = \min\{X_1, \ldots, X_k\}$. We then want to evaluate the probability $P(1 - (1 - M)^k \leq \alpha, \forall\alpha \in (0, 1)$ by evaluating the distribution of M, requiring first to isolate M.

$$P(1 - (1 - M)^k \leq \alpha) = P(1 - \alpha \leq (1 - M)^k) \overset{k\text{−th root}}{\Longrightarrow} P((1 - \alpha)^{1/k} \leq 1 - M) = P(M \leq 1 - (1 - \alpha)^{1/k})$$

We know that M is smaller than or equal to α whenever one of the X_i are smaller than or equal to α and that $X_i \perp X_j, \forall i, j \in \{1, \ldots, k\}, i \neq j$:

$$P(M \leq \alpha) = P(\exists i : X_i \leq \alpha) = 1 - P(\forall i : X_i > \alpha) = 1 - \prod_{i=1}^{k} P(X_i > \alpha)$$

Since all X_i are p-variables, i.e. $P(X_i \leq \alpha) \leq \alpha$ and $P(X_i \geq \alpha) \geq 1-\alpha$, we can see that:

$$1 - \prod_{i=1}^{k} P(X_i > \alpha) = 1 - P(X_i > \alpha)^k = 1 - (1-\alpha)^k$$

This is the probability of falsely rejecting the null hypothesis if we just select the minimum p-value among a set of p-values and do not correct it. Tippett's method then exactly cancels out this probability:

$$P(M \leq \alpha) \leq 1 - ((1-\alpha)^{1/k})^k = (1 - (1-\alpha)) = \alpha$$

Hence $\hat{p}_{\otimes,\mathtt{Tippett}}$ is a merging function.

We now consider Fisher's method. Since the X_i are p-variables and distributed uniformly as such, i.e. $X_i \sim \mathcal{U}(0,1)$, the transformations $C_i \coloneqq -2\ln(X_i)$ are χ^2_2 distributed with 2 degrees of freedom. Multiplying the independent p-variables then turns into a summation of the χ^2-distributed random variables C_i:

$$\prod_{i=1}^{k} X_i \overset{C_i = -2\ln(X_i)}{\Longrightarrow} -2\sum_{i=1}^{k} \ln C_i \eqqcolon C$$

As a sum of χ^2_2 distributed random variables, C is again χ^2-distributed with $2k$ degrees of freedom. $\hat{p}_{\otimes,\mathrm{Fisher}}$ then returns the probability of obtaining quantities at least as extreme as the observed ones:

$$\hat{p}_{\otimes,\mathrm{Fisher}} = P(\chi^2_{2k} \geq C) = 1 - F_{\chi^2_{2k}}(C)$$

resulting in a p-variable that is exactly uniformly distributed if the inputs X_i are uniformly distributed, and stochastically dominates $\mathcal{U}(0,1)$ if the inputs stochastically dominate $\mathcal{U}(0,1)$, meaning that $P(\hat{p}_{\otimes,\mathrm{Fisher}}(\boldsymbol{x}) \leq \alpha) \leq \alpha$ and hence Fisher's method is a merging function. □

$$\hat{p}_{\otimes,\mathrm{Bonferroni}}(\boldsymbol{x}) = k \cdot \min(\hat{\boldsymbol{p}})$$

$$\hat{p}_{\otimes,\mathrm{HM}}(\boldsymbol{x}) = \frac{a_k \cdot k}{\sum_{i=1}^{k} \hat{p}_i^{-1}}$$

with a_k being the correction factor depending on the number of tests to be combined:

$$a_k = \begin{cases} e \cdot \ln(2) & k = 2, \\ \frac{(y_k + k)^2}{(y_k+1)\cdot k} & k > 2. \end{cases}$$

where y_k is the unique solution to the equation

$$y^2 = k \cdot ((y+1)\cdot \ln(y+1) - y), \quad y \in (0, \infty).$$

Proposition A.4. Let $\otimes$ be a product node with k children encoding a product distribution over random variables $\boldsymbol{X}$. Let $\hat{\boldsymbol{p}} = \{\hat{p}_i\}_{i=1}^{k}$ be the p-values of the product node's children. Then $\hat{p}_{\otimes,\mathrm{Bonferroni}}$ and $\hat{p}_{\otimes,\mathrm{HM}}$ are merging functions.

Proof. We refer to (Vovk & Wang, 2020) Proposition 9 for the proof that the corrected harmonic mean computed in $\hat{p}_{\otimes,\mathrm{HM}}$ is a merging function and to Proposition 8 for the proof that the Bonferroni method computed in $\hat{p}_{\otimes,\mathrm{Bonferroni}}$ is a merging function. □

B. Visualization of Sets of Merging Functions

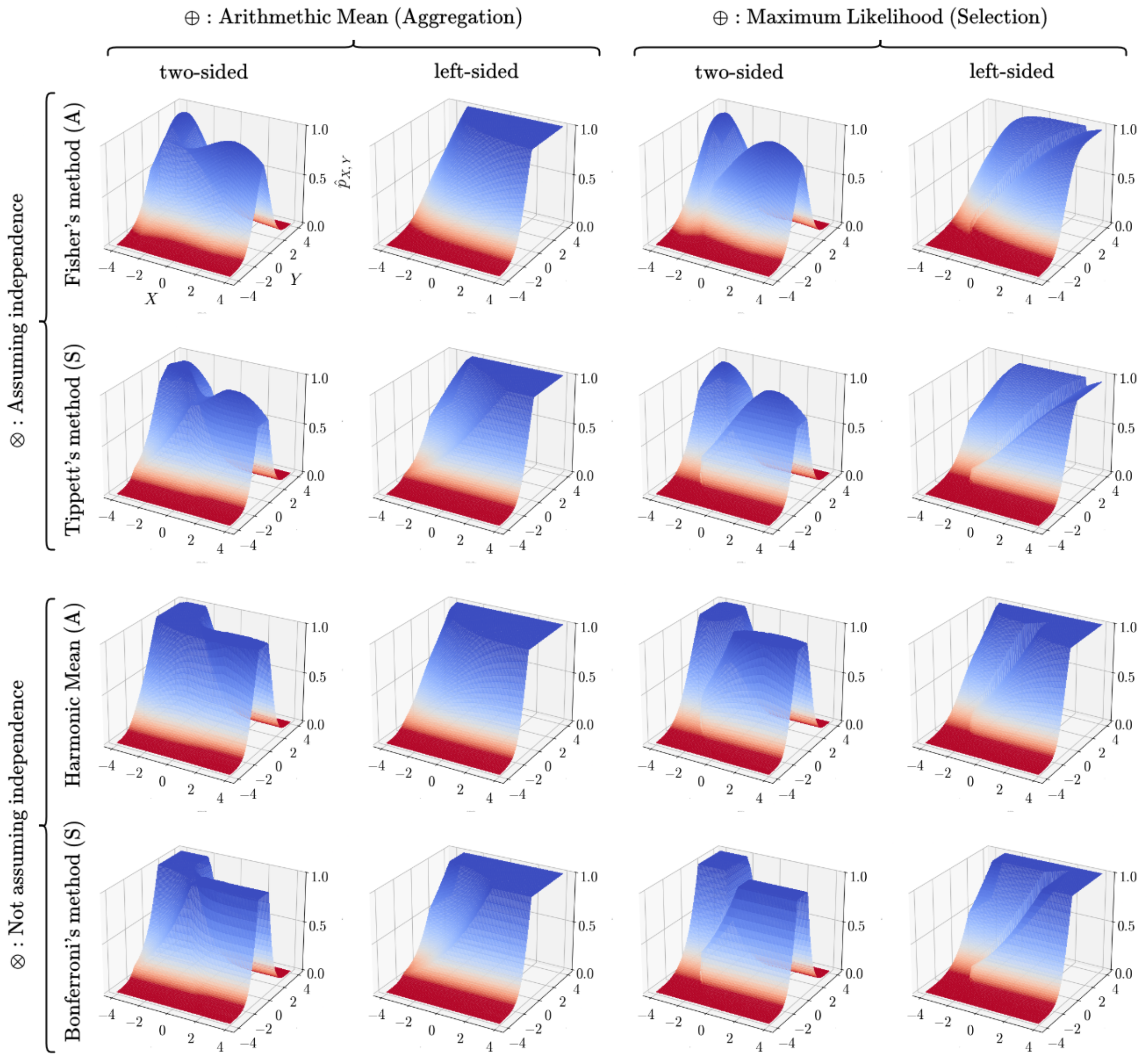

Figure 5. **Overview of merging functions in Membership Circuits:** The merging functions at sum nodes (⊕) and product nodes (⊗) influence the rejection region of the membership test. The dark-red shaded areas indicate $\hat{p}_C(\boldsymbol{x}) \leq \alpha = 0.05$ and lead to rejection of respective instances $\boldsymbol{x}$. The volumes of the rejection regions are guaranteed to be equal to or less than α (w.r.t. independence assumptions).

C. Empirical Evaluation

We present additional results and details of the experimental setups.

(Q3) MCs provide error control. We evaluated MCs, PCs (Molina et al., 2018) (sharing same architecture as MCs), and LOF (Breunig et al., 2000) on 10 common unsupervised outlier detection datasets (Goldstein & Uchida, 2016). To this end, we follow the approach of (Goldstein & Uchida, 2016) and, instead of picking the best model out of a grid search, we averaged the reported AuROC over common sets of hyper-parameters. For the circuits, we chose $m = 5\%$ of the number of observations in the train datasets as minimum to estimate the distributions in the leaf nodes and $r \in \{0.1, 0.3, 0.5\}$ as the threshold for the RDC test for independence (Lopez-Paz et al., 2013). For the LOF, we chose the number of nearest neighbors $l \in \{20, 30, 40, 50\}$. The results are summarized in Tab. 5.

	AuROC (↑, %)		
Dataset	LOF	PC	MC
aloi	74.19 $\pm$ 0.70	54.53 $\pm$ 0.79	54.32 $\pm$ 0.49
annthyroid	62.38 $\pm$ 1.04	58.83 $\pm$ 1.43	51.06 $\pm$ 2.54
b-cancer	98.06 $\pm$ 0.70	91.87 $\pm$ 0.29	96.79 $\pm$ 0.08
kdd99	46.39 $\pm$ 1.95	97.89 $\pm$ 2.84	98.66 $\pm$ 1.06
letter	86.55 $\pm$ 2.23	76.07 $\pm$ 1.46	73.87 $\pm$ 2.00
pen-global	87.25 $\pm$ 1.94	83.36 $\pm$ 1.34	93.52 $\pm$ 1.83
pen-local	98.72 $\pm$ 0.20	90.29 $\pm$ 5.67	90.30 $\pm$ 2.36
satellite	83.51 $\pm$ 12.0	90.69 $\pm$ 0.18	94.90 $\pm$ 0.25
shuttle	66.29 $\pm$ 1.69	95.34 $\pm$ 0.86	86.54 $\pm$ 16.2
speech	49.37 $\pm$ 0.87	47.53 $\pm$ 0.34	48.11 $\pm$ 0.51

Table 5. MCs and PCs demonstrate comparable theoretical performance on task of out-of-distribution detection. While the LOF, a dedicated OOD detection method, performs slightly better overall, MC and PC are expectantly close. Although MCs do not increase the AuROC in general, their membership test provides formal probabilistic guarantees (cf. Tab.3 and Fig. 4).

We have also compared our membership test with hypothesis testing baselines. To this end, we compare the FPR and the Power of MCs with fully factorized models, once assuming Gaussian distributions per random variable and once using the respective empirical CDFs, as well as Nalisnick et al. (2019)'s typicality test (conducted in identical architectures as MC). The results are presented in Tab. 6.

	MC		FF (Gaussian)		FF (Empirical)		Typicality (m=1)	
Dataset	FPR	Power	FPR	Power	FPR	Power	FPR	Power
aloi	0.96 - 1.13	3 - 5	16.9	20.3	0.1	1.3	7.1 - 15	7 - 12
annthyroid	8.72 - 9.83	9 - 18	26.5	36.8	0	0	4.7 - 55	9 - 53
b-cancer	5.60 - 8.40	40	4.5	70	0	0	12 - 15	40 - 80
kdd99	0.11 - 0.91	50 - 95	10.8	100	0	11.1	0.2 - 5.3	55 - 100
letter	1.60 - 3.27	30 - 45	1.2	5	0	0	7.1 - 12	0 - 9
pen-global	1.81 - 3.06	13 - 16	0.8	1.1	0	0	4.2 - 14	3 - 41
pen-local	2.58 - 4.81	60 - 90	1.1	0	0	0	7.7 - 17	40 - 90
satellite	4.98 - 6.23	32 - 37	0.1	42.7	0	6.7	12 - 20	63 - 69
shuttle	0 - 0.07	1 - 2	0.6	96.1	0.2	4.8	3.4 - 15	6 - 75
speech	10.3 - 32.9	11 - 26	0.5	0	0	0	N/A	N/A

Table 6. We computed p-values from fully factorized models, i.e. circuits with one root product node and one leaf per feature. "FF (Gaussian)" encodes a single univariate Gaussian in the leaves. "FF (Empirical)" computes the univariate p-values in the respective empirical CDFs. The univariate p-values are merged using the harmonic mean merging function, since we cannot assume independence between the partial test. We only report one number, as there is not randomness involved in the respective evaluations.
The Gaussian model can not approximate the data distribution well in most cases and either violates the FPR drastically or has very low power (with b-cancer and shuttle as exceptions). The empirical model respects the bound of the FPR due to the harmonic mean accounting for dependence, but is severely underpowered.

While not explicitly stated as such, fully factorized models are often assumed implicitly and, in the our experience, fairly wide-spread. We argue that the hierarchical representation of MCs allowing to model context-specific independence is a strict improvement over the factorized approach, which is also assured by the results in Tab. 6. An additional option would be constructing Union-over-Intersection tests building on the multivariate CDF of the circuit (cf. Broadrick et al. (2024)). However, the resulting test would likely have low power, since it would only test for the "outer" tails of the entire encoded distribution and e.g. miss OOD instances located between two mixture distributions. We are not aware of other instance-based hypothesis tests with analytical solutions, but welcome pointers to respective methods.

(Q4) MCs can handle missing data. Additionally to the ability of MCs to perform inference on data with missing values (compare Tab. 4), we evaluated the characteristics of the test in the respective scenarios. The experiments were conducted within the same setup as Q3 (compare above). Tab. 7 demonstrates that the membership test may still be valid when information is missing, but strongly depends on the representative quality of the underlying circuit that, in most cases, tends to decrease for the respective marginal multivariate distributions.

	PC-Testing @ $\alpha = 5\%$			
	50% MV		80% MV	
Dataset	FPR (↓, %)	Power (↑, %)	FPR (↓, %)	Power (↑, %)
aloi	2.7 - 100	4 - 100	1.2 - 100	1 - 100
annthyroid	3.7 - 100	2 - 100	2.7 - 100	35 - 100
b-cancer	1.4 - 70.9	40 - 100	0.3 - 72.3	10 - 100
kdd99	0.3 - 100	97 - 100	42 - 100	99 - 100
letter	0.4 - 19.7	7 - 38	0.3 - 20.7	3 - 27
pen-global	5.8 - 18.9	3 - 56	0 - 16	2 - 53
pen-local	3.9 - 40.2	10 - 90	0.3 - 26.1	0 - 80
satellite	4.4 - 8.7	67 - 85	2.1 - 10.3	61 - 87
shuttle	1.0 - 100	13 - 100	0.5 - 100	25 - 100
speech	0.1 - 0.5	0	0.1 - 0.6	0

Table 7. MCs are able to perform inference on missing data, but the variance of the representation quality of the underlying architectures expectantly increases when encountered with substantial information loss. While for most models, at least one test is valid, the results indicate that many models fail to correctly identify the respective marginal multivariate distributions adequately. Future work may incorporate these signals to detect mismatching sub-circuits..

D. Descriptions of Outlier Detection Datasets

Dataset	Size	Outliers	Dimensions
aloi	50,000	3.02 %	27
annthyroid	6,916	3.61 %	21
b-cancer	367	2.72 %	30
kdd99	620,098	.17 %	29
letter	1,600	6.25 %	32
pen-global	809	11.10 %	16
pen-local	6,724	.15 %	16
satellite	5,100	1.49 %	36
shuttle	46,464	1.89 %	9
speech	3,686	1.65 %	400

Table 8. Specifications of the 10 unsupervised outlier detection datasets from (Goldstein & Uchida, 2016).